AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 6
Integration and Accumulation of Change

100-Question Master Set

- Questions 1–90: Multiple Choice
- Questions 91–100: Free Response
- Complete answer key, solutions, and AP rubrics appear after Question 100

Original AP-aligned practice · Moderate level · Hand-verified

Exam Structure

Questions run continuously from 1 to 100. Every answer, worked solution, and rubric appears after Question 100.

Priority Ratings

★★★★★	**Very Important**	FTC, accumulation functions, Riemann sums, motion, average value.
★★★	**Mandatory**	Antiderivatives, definite integral evaluation, integral properties.
★★	**Important**	Basic sums, notation, and simple area computations.

- **Q1–Q90:** MCQ, four options, exactly one correct.
- **Q91–Q100:** FRQ, 9 points each; tables and graphs included.
- **Difficulty:** Easy 6 · Moderate 68 · Hard 16.
- **Calculator:** not required; FRQ 99 permits arithmetic rounding.

Topic Coverage (6.1 – 6.14)

6.1 Accumulation and Net Change

6.2 Riemann Sums (Left, Right, Midpoint)

6.3 Summation Notation

6.4 The Definite Integral

6.5 Fundamental Theorem of Calculus

6.6 Properties of Definite Integrals

6.7 FTC and Accumulated Change

6.8 Position, Velocity, Acceleration

6.9 Approximating Areas

6.10 Geometric Interpretation

6.11–6.13 Average Value

6.14 Antiderivatives and Indefinite Integrals

> **Directions:** Attempt the full paper first. For FRQs, name the theorem used and show all setups.

QUESTION 01

EASY · MULTIPLE CHOICE

★★★ MANDATORY

If $f'(t) = 3$ for all t and $f(0) = 10$, then $f(4) =$

A 30

B 18

C 22

D 12

QUESTION 02

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Water flows into a tank at $r(t) = 5 + 2t$ liters per minute. The amount added in the first 3 minutes is

A 21

B 24

C 11

D 33

QUESTION 03

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $v(t)$ is velocity, then $\int_0^8 v(t)\,dt$ gives

A its acceleration

B the particle's net change in position

C total distance traveled

D its average speed

QUESTION 04

EASY · MULTIPLE CHOICE

★★★ MANDATORY

A tank drains at a constant 3 L/min. The net change in volume over 4 minutes, $\int_0^4 (-3)\,dt$, equals

A. -12 L

B. 7 L

C. 12 L

D. -7 L

QUESTION 05

EASY · MULTIPLE CHOICE

★★★ MANDATORY

A particle has velocity $v(t) = -2$ on $[0, 5]$. Its displacement is

A. -10

B. 10

C. -2

D. 0

QUESTION 06

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A particle has velocity $v(t) = t - 2$ on $[0, 4]$. Its displacement and total distance are

A. 0 and 4

B. 4 and 0

C. 0 and 0

D. 4 and 4

QUESTION 07

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int_0^3 0\,dt =$

A 0

B Undefined

C 3

D 1

QUESTION 08

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If R(t) is the rate of rainfall in mm per hour, then $\int_0^6 R(t)\,dt$ represents

A the time of heaviest rain

B total rainfall during the first 6 hours

C the rainfall rate at $t = 6$

D average rainfall rate

QUESTION 09

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The left Riemann sum for $f(x) = x^2$ on $[0, 4]$ with $n = 2$ is

A 8

B 14

C 40

D 16

QUESTION 10

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The right Riemann sum for $f(x) = x^2$ on $[0, 4]$ with $n = 2$ is

A 32

B 8

C 20

D 40

QUESTION 11

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The midpoint Riemann sum for $f(x) = x^2$ on $[0, 4]$ with $n = 2$ is

A 24

B 10

C 20

D 28

QUESTION 12

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The left Riemann sum for $f(x) = x$ on $[0, 6]$ with $n = 3$ is

A 15

B 9

C 6

D 12

QUESTION 13

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The right Riemann sum for $f(x) = x$ on $[0, 6]$ with $n = 3$ is

A 24 B 18

C 12 D 30

QUESTION 14

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The midpoint Riemann sum for $f(x) = x$ on $[0, 6]$ with $n = 3$ is

A 27 B 18

C 9 D 21

QUESTION 15

EASY · MULTIPLE CHOICE ★★ IMPORTANT

For $f(x) = 2$ on $[1, 5]$ with $n = 4$, every Riemann sum equals

A 2 B 4

C 8 D 16

QUESTION 16

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If f is increasing on [a, b], then the left Riemann sum

A overestimates the exact area

B equals the exact area

C cannot be compared to the area

D underestimates the exact area

QUESTION 17

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If f is decreasing on [a, b], then the left Riemann sum

A is always zero

B underestimates the exact area

C overestimates the exact area

D equals the exact area

QUESTION 18

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The right Riemann sum for $f(x) = \sqrt{x}$ on $[0, 4]$ with $n = 4$ is approximately

A 6.15

B 5.00

C 7.50

D 4.00

QUESTION 19

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The midpoint Riemann sum for $f(x) = x^2$ on $[0, 2]$ with $n = 2$ is

A 3

B 2.5

C 2

D 2.67

QUESTION 20

EASY · MULTIPLE CHOICE

★★★ MANDATORY

Which Riemann sum samples f at the midpoint of each subinterval?

A Trapezoid sum

B Midpoint sum

C Right sum

D Left sum

QUESTION 21

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\Sigma(i = 1 \text{ to } 4)\ i =$

A 16

B 24

C 4

D 10

QUESTION 22

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\Sigma(i = 1 \text{ to } 5)\ (2i - 1) =$

A 15

B 30

C 25

D 20

QUESTION 23

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

$\Sigma(i = 1 \text{ to } n)\ i$ equals

A n^2

B $n(n + 1)/2$

C $n(n + 1)$

D $n^2(n + 1)/2$

QUESTION 24

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

$\Sigma(i = 1 \text{ to } n)\ i^2$ equals

A $n(n + 1)/2$

B $n^2(n + 1)^2/4$

C $n(n + 1)(2n + 1)/6$

D $n(n + 1)(n + 2)/6$

QUESTION 25

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\Sigma(i = 1 \text{ to } 4)\ 3 =$

A 24

B 3

C 7

D 12

QUESTION 26

HARD · MULTIPLE CHOICE

★★★ MANDATORY

$\Sigma(i = 1 \text{ to } 6)\ i^3 =$

A 441

B 21

C 36

D 196

QUESTION 27

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int_0^4 3\ dx =$

A 3

B 12

C 4

D 7

QUESTION 28

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_{-2}^{3} 2x\,dx =$

A 13

B 5

C 10

D 0

QUESTION 29

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int_{0}^{3} x\,dx =$

A 9

B 3

C 9/2

D 6

QUESTION 30

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_{0}^{4} (4 - x)\,dx =$

A 16

B 4

C 12

D 8

QUESTION 31

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_{-3}^{3} |x|\, dx =$

A 18

B 9

C 4.5

D 0

QUESTION 32

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int_{0}^{2} (x + 1)\, dx =$

A 6

B 3

C 2

D 4

QUESTION 33

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int_{1}^{4} 1\, dx =$

A 4

B 3

C 5

D 1

QUESTION 34

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_{-1}^{1} x^3\,dx =$

A -2

B $1/2$

C 0

D 2

QUESTION 35

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int 5\,dx =$

A $5 + C$

B $5x$

C $x^5 + C$

D $5x + C$

QUESTION 36

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int x^4\,dx =$

A $x^3/3 + C$

B $4x^3 + C$

C $x^5/5 + C$

D $x^5 + C$

QUESTION 37

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int 3x^2\,dx =$

A $3x^3 + C$

B $6x + C$

C $x^3 + C$

D $x^2 + C$

QUESTION 38

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int (2x + 3)\,dx =$

A $2x^2 + 3x + C$

B $2 + C$

C $x^2 + 3 + C$

D $x^2 + 3x + C$

QUESTION 39

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int x^{-2}\,dx =$

A $\ln|x| + C$

B $x^{-1} + C$

C $-2x^{-3} + C$

D $-x^{-1} + C$

QUESTION 40

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int \sqrt{x}\, dx =$

A (1/2)x^(1/2) + C

B (3/2)x^(3/2) + C

C (2/3)x^(3/2) + C

D x^(3/2) + C

QUESTION 41

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\int (1/x)\, dx =$

A $\ln|x| + C$

B $1/x^2 + C$

C $x \ln x + C$

D $e^x + C$

QUESTION 42

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\int e^x\, dx =$

A $e^{x+1} + C$

B $xe^x + C$

C $\ln x + C$

D $e^x + C$

QUESTION 43

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\int \cos x \, dx =$

A $\sec^2 x + C$

B $\sin x + C$

C $-\cos x + C$

D $-\sin x + C$

QUESTION 44

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\int \sin x \, dx =$

A $-\cos x + C$

B $\cos x + C$

C $\sin x + C$

D $-\sin x + C$

QUESTION 45

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int \sec^2 x \, dx =$

A $\tan x + C$

B $-\tan x + C$

C $\sec x + C$

D $2 \sec x \tan x + C$

QUESTION 46

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\int (x^2 - 4x + 1)\,dx =$

A $x^3/3 - 2x^2 + x + C$

B $x^3 - 4x^2 + x + C$

C $3x - 4 + C$

D $x^3/3 - 4x^2 + x + C$

QUESTION 47

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\int_1^3 (2x + 1)\,dx =$

A 12

B 8

C 16

D 10

QUESTION 48

EASY · MULTIPLE CHOICE ★★ IMPORTANT

$\int_0^2 3x^2\,dx =$

A 6

B 12

C 8

D 4

QUESTION 49

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\int_0^{\pi} \sin x \, dx =$

A π

B 2

C 0

D 1

QUESTION 50

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\int_1^{e} (1/x) \, dx =$

A 1

B 0

C $\ln e$

D e

QUESTION 51

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_0^{1} e^x \, dx =$

A 1

B e

C $e - 1$

D $e + 1$

QUESTION 52

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_2^5 (x^2 - 1)\,dx =$

A 36

B 28

C 42

D 30

QUESTION 53

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_{-1}^1 (x^2 + 1)\,dx =$

A 4/3

B 2

C 4

D 8/3

QUESTION 54

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_0^4 \sqrt{x}\,dx =$

A 4

B 16/3

C 8/3

D 8

QUESTION 55

EASY · MULTIPLE CHOICE ★★ IMPORTANT

$\int_0^2 (2x + 3)\,dx =$

A 10
B 14
C 6
D 7

QUESTION 56

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\int_0^3 (2x - 4)\,dx =$

A -6
B 0
C 3
D -3

QUESTION 57

EASY · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $F'(x) = f(x)$, then $\int_a^b f(x)\,dx$ equals

A $F(a) - F(b)$
B $f(b) - f(a)$
C $F'(b) - F'(a)$
D $F(b) - F(a)$

QUESTION 58

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\int_0^2 (x^2 + 2)\,dx =$

A 20/3
B 22/3
C 14/3
D 8

QUESTION 59

EASY · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Which property states $\int_0^5 f + \int_5^8 f = \int_0^8 f$?

A Comparison property
B Reversing limits
C Additivity over intervals
D Constant multiple

QUESTION 60

EASY · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Reversing the limits of a definite integral gives

A the reciprocal
B the same value
C the negative of the value
D zero

QUESTION 61

EASY · MULTIPLE CHOICE ★★★ MANDATORY

$\int_0^4 5f(x)\,dx$ equals

A $5\int_0^4 f(x)\,dx$

B $5 + \int_0^4 f(x)\,dx$

C $\int_0^4 f(x)\,dx + 5$

D $\int_0^{20} f(x)\,dx$

QUESTION 62

EASY · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $f(x) \geq 0$ on $[a, b]$, then

A $\int_a^b f(x)\,dx \leq 0$

B $\int_a^b f(x)\,dx = 0$

C $\int_a^b f(x)\,dx \geq 0$

D $\int_a^b f(x)\,dx = b - a$

QUESTION 63

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\int_{-2}^2 x^3\,dx =$

A 0

B 8

C 4

D 16

QUESTION 64

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\int_{-3}^{3} x^2\,dx$ equals

A 27

B 18

C 9

D 0

QUESTION 65

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $f(x) \le g(x)$ on $[a, b]$, then

A $\int f \le \int g$

B $\int f \ge \int g$

C no comparison is possible

D $\int f = \int g$

QUESTION 66

EASY · MULTIPLE CHOICE ★★ IMPORTANT

$\int_{0}^{2} 4\,dx =$

A 2

B 8

C 6

D 4

QUESTION 67

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $F(x) = \int_0^x (t + 1)\,dt$, then $F'(x) =$

A $x + 1$

B 1

C $x^2/2 + x$

D x

QUESTION 68

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $G(x) = \int_1^x \cos t\,dt$, then $G'(x) =$

A $\cos 1$

B $\sin x$

C $\cos x$

D $-\sin x$

QUESTION 69

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $H(x) = \int_0^{(x^2)} \sin t\,dt$, then $H'(x) =$

A $2x \sin(x^2)$

B $\sin(x^2)$

C $x^2 \cos(x^2)$

D $\sin(x^2) \cdot 2x + C$

QUESTION 70

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $F(x) = \int_2^x 3t^2\,dt$, then $F'(2) =$

A 12
B 4
C 8
D 0

QUESTION 71

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $F(x) = \int_0^x 2t\,dt$, then $F(3) =$

A 6
B 3
C 9
D 18

QUESTION 72

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $A(x) = \int_1^x (1/t)\,dt$, then $A(e) =$

A 0
B e
C 1/e
D 1

QUESTION 73

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $F(x) = \int_0^x (t^2 - 4)\,dt$, the critical numbers of F are

A $x = 0$ only

B $x = 2$ only

C none

D $x = -2$ and $x = 2$

QUESTION 74

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $G(x) = \int_0^x (2 - t)\,dt$, then G is increasing when

A $x > 2$

B $x > 0$

C $x < 2$

D $x < 0$

QUESTION 75

HARD · MULTIPLE CHOICE ★★★ MANDATORY

If $F(x) = \int_{-1}^x t\,dt$, then $F(2) =$

A 2

B 4

C 1/2

D 3/2

QUESTION 76

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \int_0^{(x^3)} e^t\ dt\ ] =$

A $3x^2\ e^{(x^3)}$

B $e^{(x^3)}$

C $e^{(x^3)} \cdot x^3$

D $e^{(3x^2)}$

QUESTION 77

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The average value of $f(x) = x^2$ on $[0, 3]$ is

A 9

B 9/2

C 3

D 6

QUESTION 78

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The average value of $f(x) = 2x + 1$ on $[0, 4]$ is

A 10

B 9

C 5

D 4

QUESTION 79

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The average value of $f(x) = \sin x$ on $[0, \pi]$ is

A $1/\pi$ B 2

C 0 D $2/\pi$

QUESTION 80

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The average value of $f(x) = e^x$ on $[0, 1]$ is

A 1 B e

C $e/2$ D $e - 1$

QUESTION 81

EASY · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $\int_0^6 f(x)\,dx = 18$, then the average value of f on $[0, 6]$ is

A $1/3$ B 6

C 3 D 18

QUESTION 82

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The average value of $f(x) = 4 - x$ on $[0, 4]$ is

A 0
B 1
C 2
D 4

QUESTION 83

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $v(t) = 3t^2$, the displacement on $[0, 2]$ is

A 4
B 8
C 12
D 6

QUESTION 84

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $v(t) = 2t - 4$, the displacement on $[0, 6]$ is

A 0
B 20
C 12
D 36

QUESTION 85

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $v(t) = 2t - 4$ on $[0, 6]$, the total distance traveled is

A 24
B 20
C 16
D 12

QUESTION 86

EASY · MULTIPLE CHOICE ★★★ MANDATORY

If $v(t) = 4$ on $[1, 5]$, the distance traveled is

A 16
B 8
C 20
D 4

QUESTION 87

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A particle has acceleration $a(t) = 6$ and initial velocity $v(0) = 2$. Then $v(3) =$

A 6
B 20
C 20/3
D 18

QUESTION 88

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If velocity is always positive on [a, b], then

A total distance exceeds displacement

B distance is zero

C displacement is zero

D total distance equals displacement

QUESTION 89

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $s'(t) = v(t)$ and $s(0)$ is known, then $s(t) =$

A $s(0) + v(t)$

B $v(0) + \int_0^t v(x)\,dx$

C $\int_0^t v(x)\,dx$ only

D $s(0) + \int_0^t v(x)\,dx$

QUESTION 90

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $\int_0^5 |v(t)|\,dt = 30$ and $\int_0^5 v(t)\,dt = 12$, then over $[0, 5]$ the particle

A traveled 12 units and ended 30 units away

B ended 18 units from its start

C traveled 42 units

D traveled 30 units and ended 12 units from its start

Question 91: Riemann Sums from a Table

★★★★★ Very Important · 9 points

The function f is continuous on [0, 8]. Selected values are given in the table.

x	0	2	4	6	8
f(x)	3	5	4	2	1

Use the listed table values to evaluate the approximations below.

(a) [3 points] Use a left Riemann sum with n = 4 to estimate $\int_0^8 f(x)\,dx$.

(b) [3 points] Use a right Riemann sum with n = 4 to estimate the same integral.

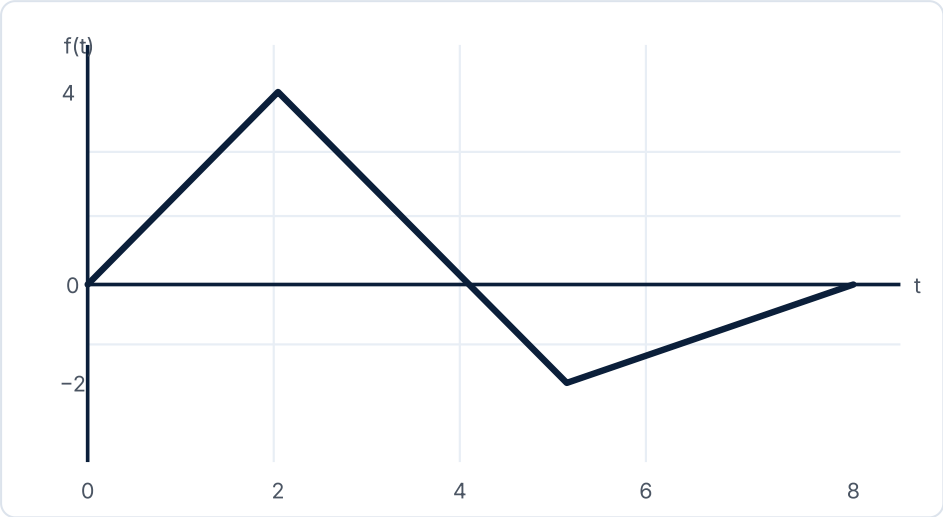
(c) [3 points] Use the trapezoid rule with n = 4 and explain why it is the average of your two earlier estimates.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 92: Definite Integrals from a Graph

★★★★★ Very Important | 9 points

The graph of f consists of straight line segments and is shown for $0 \le t \le 8$.



Graph of f on $0 \le t \le 8$, made of straight line segments.

(a) [3 points] Find $\int_0^8 f(t)\,dt$. Explain how the signs of the regions affect your answer.

(b) [3 points] Find the total area between the graph of f and the t-axis on [0, 8].

(c) [2 points] Find the average value of f on [0, 8].

(d) [1 points] Explain why your answers to (a) and (b) are different numbers.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 93: Analyzing an Accumulation Function

★★★★★ Very Important | 9 points

Let $F(x) = \int_0^x (t^2 - 4)\,dt$.

(a) [2 points] Find $F'(x)$ and state the theorem you used.

(b) [3 points] Find an explicit formula for $F(x)$.

(c) [2 points] Find the critical numbers of F and classify each as a local maximum or local minimum.

(d) [2 points] Evaluate $F(2)$ and $F(-2)$.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 94: FTC and Average Value

★★★★★ Very Important | 9 points

Let $f(x) = 3x^2 - 2x$.

(a) [3 points] Evaluate $\int_1^4 f(x)\,dx$.

(b) [2 points] Find the average value of f on [1, 4].

(c) [4 points] Find the value c in [1, 4] such that f(c) equals the average value from part (b).

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 95: Motion: Displacement and Distance

★★★★★ Very Important | 9 points

A particle moves along a line with velocity $v(t) = t^2 - 4t + 3$ for $0 \le t \le 5$, where v is in m/s.

(a) [3 points] Find the displacement over [0, 5].

(b) [4 points] Find the total distance traveled over [0, 5].

(c) [2 points] During which time intervals is the particle moving in the positive direction?

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 96: From a Riemann Sum to an Integral

★★★★★ Very Important · 9 points

Consider $\lim_{n\to\infty} \sum_{i=1}^{n} (2 + 6i/n)^2 \cdot (6/n)$.

(a) [3 points] Identify Δx and the sample points x_i, and state the interval $[a, b]$.

(b) [3 points] Write the limit as a definite integral and identify the integrand $f(x)$.

(c) [3 points] Evaluate the integral exactly.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 97: Integral of a Piecewise Function

★★★ Mandatory 9 points

Let $f(x) = x + 1$ for $0 \le x \le 2$, and $f(x) = 5 - x$ for $2 < x \le 5$.

(a) [3 points] Evaluate $\int_0^5 f(x)\,dx$.

(b) [2 points] Find the average value of f on [0, 5].

(c) [4 points] Determine whether there is a value c in [0, 5] with f(c) equal to the average value. If so, find it.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 98: Accumulation with a Variable Upper Limit

★★★★★ Very Important | 9 points

Let $G(x) = \int_1^{(x^2)} (t + 1)\, dt$.

(a) [3 points] Find $G'(x)$.

(b) [2 points] Find an explicit formula for $G(x)$ and use it to confirm your answer to part (a).

(c) [2 points] Evaluate $G(2)$.

(d) [2 points] Find all critical numbers of G.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 99: Applied Accumulation from a Table

★★★★★ Very Important | 9 points

Water flows into a tank. The rate R(t), in liters per minute, is given at selected times.

t (min)	0	2	4	6	8
R(t) (L/min)	5	7	6	3	2

(a) [3 points] Use a trapezoid approximation with n = 4 to estimate the total amount of water added on [0, 8].

(b) [2 points] Compute the left and right Riemann sums with n = 4.

(c) [2 points] Show that the trapezoid estimate is the average of the left and right sums.

(d) [2 points] Interpret your answer from part (a) in the context of the problem, with units.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Question 100: Net Area versus Total Area

★★★★★ Very Important · 9 points

Let $f(x) = x - 2$ on the interval $[0, 5]$.

(a) [2 points] Evaluate $\int_0^5 f(x)\,dx$.

(b) [3 points] Find the total area between the graph of f and the x-axis on [0, 5].

(c) [2 points] Find the average value of f on [0, 5].

(d) [2 points] Explain why your answers to (a) and (b) differ.

Show your work. Name the theorem or rule used, show every setup, and include units where the context requires them.

Questions 1–100 Complete

Stop here if you are working under timed conditions. The next pages contain the answer key, worked MCQ solutions, and FRQ model solutions with rubrics.

> **Scoring for this practice set:** 90 MCQ × 1 point + 10 FRQ × 9 points = 180 raw points. This is a practice scheme, not an official AP conversion.

Multiple-Choice Answer Key

Letters reflect the shuffled option order used in this document.

Q01 C ★★★	Q02 B ★★★★★	Q03 B ★★★★★	Q04 A ★★★	Q05 A ★★★	Q06 A ★★★★★	Q07 A ★★	Q08 B ★★★★★	Q09 A ★★★★★	Q10 D ★★★★★
Q11 C ★★★★★	Q12 D ★★★	Q13 A ★★★	Q14 B ★★★	Q15 C ★★	Q16 D ★★★★★	Q17 C ★★★★★	Q18 A ★★★	Q19 B ★★★	Q20 B ★★★
Q21 D ★★	Q22 C ★★★	Q23 B ★★★★★	Q24 C ★★★★★	Q25 D ★★	Q26 A ★★★	Q27 B ★★	Q28 B ★★★	Q29 C ★★	Q30 D ★★★
Q31 B ★★★	Q32 D ★★	Q33 B ★★	Q34 C ★★★	Q35 D ★★	Q36 C ★★	Q37 C ★★	Q38 D ★★	Q39 D ★★★	Q40 C ★★★
Q41 A ★★★★★	Q42 D ★★★★★	Q43 B ★★★★★	Q44 A ★★★★★	Q45 A ★★★	Q46 A ★★★	Q47 D ★★★	Q48 C ★★	Q49 B ★★★★★	Q50 A ★★★★★
Q51 C ★★★	Q52 A ★★★	Q53 D ★★★	Q54 B ★★★	Q55 A ★★	Q56 D ★★★	Q57 D ★★★★★	Q58 A ★★★	Q59 C ★★★★★	Q60 C ★★★★★
Q61 A ★★★	Q62 C ★★★★★	Q63 A ★★★	Q64 B ★★★	Q65 A ★★★	Q66 B ★★	Q67 A ★★★★★	Q68 C ★★★★★	Q69 A ★★★★★	Q70 A ★★★
Q71 C ★★★	Q72 D ★★★★★	Q73 D ★★★★★	Q74 C ★★★★★	Q75 D ★★★	Q76 A ★★★★★	Q77 C ★★★★★	Q78 C ★★★	Q79 D ★★★★★	Q80 D ★★★
Q81 C ★★★★★	Q82 C ★★★	Q83 B ★★★★★	Q84 C ★★★★★	Q85 B ★★★★★	Q86 A ★★★	Q87 B ★★★★★	Q88 D ★★★★★	Q89 D ★★★★★	Q90 D ★★★★★

Recheck tip: For any question you missed, read the matching solution page — it explains why each wrong option is wrong, which is where most errors come from.

Question 01

★★★ Mandatory

Correct answer: (C) 22

STEP-BY-STEP SOLUTION

Net change = $\int_0^4 3\,dt = 12$. So $f(4) = f(0) + 12 = 10 + 12 = 22$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This double-counts the initial value.

✗ **(B)** This adds $10 + 2 \cdot 4$ instead of $10 + 3 \cdot 4$.

✗ **(D)** This is only the accumulated change and ignores the starting value 10.

Question 02

★★★★★ Very Important

Correct answer: (B) 24

STEP-BY-STEP SOLUTION

$\int_0^3 (5 + 2t)\,dt = 5t + t^2 \big|_0^3 = 15 + 9 = 24$ liters.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $5 \cdot 3 + 2 \cdot 3$, treating $2t$ as a constant.

✗ **(C)** This is $r(3)$, the rate at one instant, not the total.

✗ **(D)** Integrates $2t$ as $2t^2$, giving $15 + 18 = 33$.

Question 03

★★★★★ Very Important

Correct answer: (B) the particle's net change in position

STEP-BY-STEP SOLUTION

The integral of velocity is displacement, that is, net change in position.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Acceleration is v'.

✗ **(C)** Total distance requires $\int |v|\,dt$.

✗ **(D)** Average speed is total distance divided by time.

Question 04

★★★ Mandatory

Correct answer: (A) −12 L

STEP-BY-STEP SOLUTION

The rate is negative because volume is decreasing: $(-3)(4) = -12$ liters.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Both ignores the sign and adds.

✗ **(C)** Ignores that the outflow rate is negative.

✗ **(D)** Adds instead of multiplying.

Question 05

★★★ Mandatory

Correct answer: (A) −10

STEP-BY-STEP SOLUTION

Displacement = $\int_0^5 (-2)\,dt = -2(5) = -10$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is total distance, $|\text{displacement}|$.

✗ **(C)** This is the velocity value, not the displacement.

✗ **(D)** The velocity is nonzero, so displacement is nonzero.

Question 06

★★★★★ Very Important

Correct answer: (A) 0 and 4

STEP-BY-STEP SOLUTION

Displacement = $\int_0^4 (t-2)\,dt = [t^2/2 - 2t]_0^4 = 8 - 8 = 0$. Distance = $\int_0^2 (2-t)\,dt + \int_2^4 (t-2)\,dt = 2 + 2 = 4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Reverses displacement and distance.

✗ **(C)** The particle does move, so distance is not 0.

✗ **(D)** Displacement is 0, not 4.

Question 07

★★ Important

Correct answer: (A) 0

STEP-BY-STEP SOLUTION

The integrand is identically zero, so the accumulated change is 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** The integral of 0 is well defined.

✗ **(C)** Confuses the interval length with the value.

✗ **(D)** No basis.

Question 08

★★★★★ Very Important

Correct answer: (B) total rainfall during the first 6 hours

STEP-BY-STEP SOLUTION

Integrating a rate over a time interval gives the total accumulated amount.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The integral gives a total, not a time.

✗ **(C)** That is R(6), not the integral.

✗ **(D)** Average rate would divide the integral by 6.

Question 09

★★★★★ Very Important

Correct answer: (A) 8

STEP-BY-STEP SOLUTION

$\Delta x = 2$. Left endpoints are $x = 0$ and $x = 2$, so $L_2 = 2(f(0) + f(2)) = 2(0 + 4) = 8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Mixes a left endpoint with a right endpoint.

✗ **(C)** This is the right Riemann sum.

✗ **(D)** Uses only $f(4) = 16$.

Question 10

★★★★★ Very Important

Correct answer: (D) 40

STEP-BY-STEP SOLUTION

$\Delta x = 2$. Right endpoints are $x = 2$ and $x = 4$, so $R_2 = 2(f(2) + f(4)) = 2(4 + 16) = 40$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles the left sum incorrectly.

✗ **(B)** This is the left Riemann sum.

✗ **(C)** Uses $2(4 + 6)$ with an arithmetic slip.

Question 11

★★★★★ Very Important

Correct answer: (C) 20

STEP-BY-STEP SOLUTION

$\Delta x = 2$. Midpoints are $x = 1$ and $x = 3$, so $M_2 = 2(f(1) + f(3)) = 2(1 + 9) = 20$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses midpoints 1 and 2.

✗ **(B)** Forgets to multiply by $\Delta x = 2$.

✗ **(D)** Uses midpoints 1 and 4.

Question 12

★★★ Mandatory

Correct answer: (D) 12

STEP-BY-STEP SOLUTION

$\Delta x = 2$. Left endpoints 0, 2, 4 give $L_3 = 2(0 + 2 + 4) = 12$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses right endpoints 2, 4, 6 but forgets Δx.

✗ **(B)** Forgets Δx.

✗ **(C)** Uses only the last two terms.

Question 13

★★★ Mandatory

Correct answer: (A) 24

STEP-BY-STEP SOLUTION

Right endpoints 2, 4, 6 give $R_3 = 2(2 + 4 + 6) = 24$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses midpoints 1, 3, 5 but forgets Δx.

✗ **(C)** This is the left Riemann sum.

✗ **(D)** Adds an extra term.

Question 14

★★★ Mandatory

Correct answer: (B) 18

STEP-BY-STEP SOLUTION

Midpoints 1, 3, 5 give $M_3 = 2(1 + 3 + 5) = 18$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses endpoints 1, 3, 9.

✗ **(C)** Forgets $\Delta x = 2$.

✗ **(D)** Uses midpoints 1, 3, 6.5.

Question 15

★★ Important

Correct answer: (C) 8

STEP-BY-STEP SOLUTION

$\Delta x = 1$ and $f = 2$ at every sample point, so any sum is $4 \cdot 1 \cdot 2 = 8$, which is also the exact area.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the function value.

✗ **(B)** Uses only $\Delta x \cdot f$ with one rectangle.

✗ **(D)** Uses 4 rectangles of height 4.

Question 16

★★★★★ Very Important

Correct answer: (D) underestimates the exact area

STEP-BY-STEP SOLUTION

On each subinterval the left endpoint gives the smallest value of an increasing function, so the rectangles fall short.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** That describes the right Riemann sum for an increasing function.

✗ **(B)** Only true for a constant function.

✗ **(C)** The comparison is always possible for monotonic functions.

Question 17

★★★★★ Very Important

Correct answer: (C) overestimates the exact area

STEP-BY-STEP SOLUTION

On each subinterval a decreasing function is largest at its left endpoint, so the rectangles rise above the curve.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The function need not average to zero.

✗ **(B)** That is the right Riemann sum for a decreasing function.

✗ **(D)** Only true for a constant function.

Question 18

★★★ Mandatory

Correct answer: (A) 6.15

STEP-BY-STEP SOLUTION

$\Delta x = 1$ and right endpoints are 1, 2, 3, 4. The sum is $\sqrt{1} + \sqrt{2} + \sqrt{3} + \sqrt{4} \approx 1 + 1.414 + 1.732 + 2 = 6.146$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses left endpoints 0, 1, 2, 3, giving about 4.15.

✗ **(C)** Uses midpoints and adds an extra unit.

✗ **(D)** Uses only the last term doubled.

Question 19

★★★ Mandatory

Correct answer: (B) 2.5

STEP-BY-STEP SOLUTION

$\Delta x = 1$ and midpoints are 0.5 and 1.5. The sum is $0.25 + 2.25 = 2.5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses endpoints 1 and 2.

✗ **(C)** Uses endpoints 0 and 2.

✗ **(D)** This is the exact integral value 8/3.

Question 20

★★★ Mandatory

Correct answer: (B) Midpoint sum

STEP-BY-STEP SOLUTION

The midpoint sum evaluates f at the center of every subinterval.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Averages left and right endpoint values.

✗ **(C)** Uses right endpoints.

✗ **(D)** Uses left endpoints.

Question 21

★★ Important

Correct answer: (D) 10

STEP-BY-STEP SOLUTION

$1 + 2 + 3 + 4 = 10$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is 4^2.

✗ **(B)** This is $4!$.

✗ **(C)** Counts only the last term.

Question 22

★★★ Mandatory

Correct answer: (C) 25

STEP-BY-STEP SOLUTION

The terms are 1 + 3 + 5 + 7 + 9 = 25.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sums only 1 + 2 + 3 + 4 + 5.

✗ **(B)** Adds an extra term.

✗ **(D)** Forgets the −1 in each term partially.

Question 23

★★★★★ Very Important

Correct answer: (B) n(n + 1)/2

STEP-BY-STEP SOLUTION

The sum of the first n positive integers is n(n + 1)/2.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the sum of the first n odd numbers.

✗ **(C)** Misses the factor 1/2.

✗ **(D)** This resembles the sum of squares formula.

Question 24

★★★★★ Very Important

Correct answer: (C) n(n + 1)(2n + 1)/6

STEP-BY-STEP SOLUTION

The standard formula for the sum of squares is n(n + 1)(2n + 1)/6.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the sum of the first n integers.

✗ **(B)** This is the sum of the first n cubes.

✗ **(D)** This is a binomial-coefficient form.

Question 25

★★ Important

Correct answer: (D) 12

STEP-BY-STEP SOLUTION

A constant 3 added four times gives 4·3 = 12.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies by 8 instead of 4.

✗ **(B)** Counts only one term.

✗ **(C)** Adds the index to the constant.

Question 26

★★★ Mandatory

Correct answer: (A) 441

STEP-BY-STEP SOLUTION

The sum of the first n cubes is $[n(n + 1)/2]^2$. Here $[6 \cdot 7/2]^2 = 21^2 = 441$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is the sum 1 + 2 + ... + 6.

✗ **(C)** This is 6^2.

✗ **(D)** This is 14^2, using n(n+1) instead.

Question 27

★★ Important

Correct answer: (B) 12

STEP-BY-STEP SOLUTION

The integral is the area of a rectangle of height 3 and width 4, which is 12.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the height.

✗ **(C)** Uses only the width.

✗ **(D)** Adds height and width.

Question 28

★★★ Mandatory

Correct answer: (B) 5

STEP-BY-STEP SOLUTION

An antiderivative is x^2. Evaluating gives $3^2 - (-2)^2 = 9 - 4 = 5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds 9 and 4.

✗ **(C)** Forgets to subtract the lower-limit value.

✗ **(D)** Treats the odd integrand over a symmetric-looking interval as zero, but the interval is not symmetric.

Question 29

★★ Important

Correct answer: (C) 9/2

STEP-BY-STEP SOLUTION

An antiderivative is $x^2/2$. Evaluating gives $9/2 - 0 = 9/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by 2.

✗ **(B)** Uses x instead of $x^2/2$.

✗ **(D)** Uses $3 \cdot 2$.

Question 30

★★★ Mandatory

Correct answer: (D) 8

STEP-BY-STEP SOLUTION

The region is a triangle with base 4 and height 4, so the area is $(1/2)(4)(4) = 8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 4^2.

✗ **(B)** Uses base times height without 1/2.

✗ **(C)** Adds the base and height incorrectly.

Question 31

★★★ Mandatory

Correct answer: (B) 9

STEP-BY-STEP SOLUTION

The graph is two congruent triangles, each with base 3 and height 3. Total area = $2 \cdot (1/2)(3)(3) = 9$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles the correct area.

✗ **(C)** Uses only one triangle.

✗ **(D)** Treats $|x|$ as an odd function; it is even.

Question 32

★★ Important

Correct answer: (D) 4

STEP-BY-STEP SOLUTION

An antiderivative is $x^2/2 + x$. Evaluating gives $(2 + 2) - 0 = 4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the constant term's contribution is only 2, not 4.

✗ **(B)** Uses only the triangle area.

✗ **(C)** Uses only the linear part.

Question 33

★★ Important

Correct answer: (B) 3

STEP-BY-STEP SOLUTION

The integral of 1 over an interval is the interval length: $4 - 1 = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the upper limit.

✗ **(C)** Adds the limits.

✗ **(D)** Uses the integrand value.

Question 34

★★★ Mandatory

Correct answer: (C) 0

STEP-BY-STEP SOLUTION

x^3 is an odd function and the interval is symmetric about 0, so the integral is 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes only the left half with a sign.

✗ **(B)** Uses the average value.

✗ **(D)** Computes only the right half.

Question 35

★★ Important

Correct answer: (D) 5x + C

STEP-BY-STEP SOLUTION

The antiderivative of a constant k is kx + C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the variable x.

✗ **(B)** Omits the constant of integration, which is required for an indefinite integral.

✗ **(C)** Treats 5 as an exponent.

Question 36

★★ Important

Correct answer: (C) $x^5/5 + C$

STEP-BY-STEP SOLUTION

The power rule gives x^(4+1)/(4 + 1) + C = $x^5/5$ + C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the wrong exponent.

✗ **(B)** Applies the power rule as if differentiating.

✗ **(D)** Forgets to divide by the new exponent.

Question 37

★★ Important

Correct answer: (C) $x^3 + C$

STEP-BY-STEP SOLUTION

Pull out the 3 and apply the power rule: $3 \cdot (x^3/3) + C = x^3 + C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by the new exponent.

✗ **(B)** Differentiates instead of integrating.

✗ **(D)** Reduces the exponent incorrectly.

Question 38

★★ Important

Correct answer: (D) $x^2 + 3x + C$

STEP-BY-STEP SOLUTION

Integrate term by term: $2x \rightarrow x^2$ and $3 \rightarrow 3x$, then add C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Fails to divide the 2x term by 2.

✗ **(B)** Differentiates instead.

✗ **(C)** Treats 3 as $3x^2/2$.

Question 39

★★★ Mandatory

Correct answer: (D) $-x^{-1} + C$

STEP-BY-STEP SOLUTION

The power rule gives $x^{-1}/(-1) + C = -1/x + C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** That is the antiderivative of 1/x, not x^{-2}.

✗ **(B)** Drops the negative sign from dividing by −1.

✗ **(C)** Applies the power rule as if differentiating.

Question 40

★★★ Mandatory

Correct answer: (C) (2/3)x^(3/2) + C

STEP-BY-STEP SOLUTION

Write √x = x^(1/2). The power rule gives x^(3/2)/(3/2) + C = (2/3)x^(3/2) + C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiates instead.

✗ **(B)** Inverts the fraction 2/3.

✗ **(D)** Forgets to divide by 3/2.

Question 41

★★★★★ Very Important

Correct answer: (A) ln|x| + C

STEP-BY-STEP SOLUTION

The natural logarithm is the antiderivative of 1/x; absolute values are needed because the domain excludes 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Differentiates instead.

✗ **(C)** Confuses with the antiderivative of ln x.

✗ **(D)** Confuses with the derivative of e^x.

Question 42

★★★★★ Very Important

Correct answer: (D) e^x + C

STEP-BY-STEP SOLUTION

The exponential function is its own antiderivative.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds 1 to the exponent.

✗ **(B)** Applies a product rule idea that does not apply here.

✗ **(C)** Confuses with the antiderivative of 1/x.

Question 43

★★★★★ Very Important

Correct answer: (B) sin x + C

STEP-BY-STEP SOLUTION

Since d/dx(sin x) = cos x, the antiderivative of cos x is sin x + C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** That is the antiderivative of tan x's related derivative.

✗ **(C)** This is the antiderivative of sin x with the wrong sign.

✗ **(D)** This is the antiderivative of −cos x, or the derivative relationship of sin.

Question 44

★★★★★ Very Important

Correct answer: (A) −cos x + C

STEP-BY-STEP SOLUTION

Since d/dx(−cos x) = sin x, the antiderivative is −cos x + C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Has the wrong sign.

✗ **(C)** Is the original function.

✗ **(D)** Is the negative of the integrand.

Question 45

★★★ Mandatory

Correct answer: (A) tan x + C

STEP-BY-STEP SOLUTION

Because d/dx(tan x) = $\sec^2 x$, the antiderivative is tan x + C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Has the wrong sign.

✗ **(C)** Differentiates the wrong trig function.

✗ **(D)** That is the derivative of sec x.

Question 46

★★★ Mandatory

Correct answer: (A) $x^3/3 - 2x^2 + x + C$

STEP-BY-STEP SOLUTION

Integrate term by term: $x^2 \to x^3/3$, $-4x \to -2x^2$, $1 \to x$, then add C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide x^2 by 3.

✗ **(C)** Differentiates instead.

✗ **(D)** Fails to divide $-4x$ by 2.

Question 47

★★★ Mandatory

Correct answer: (D) 10

STEP-BY-STEP SOLUTION

An antiderivative is $x^2 + x$. Evaluating gives $(9 + 3) - (1 + 1) = 12 - 2 = 10$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates only at the upper limit.

✗ **(B)** Forgets the $+x$ term.

✗ **(C)** Uses $x^2 + x$ at 3 but forgets to subtract.

Question 48

★★ Important

Correct answer: (C) 8

STEP-BY-STEP SOLUTION

An antiderivative is x^3. Evaluating gives $8 - 0 = 8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $3x^2$ evaluated at 2 and forgets to integrate fully.

✗ **(B)** Multiplies 3 by 2^2 without integrating.

✗ **(D)** Uses x^2 instead of x^3.

Question 49

★★★★★ Very Important

Correct answer: (B) 2

STEP-BY-STEP SOLUTION

An antiderivative is $-\cos x$. Evaluating gives $-\cos \pi + \cos 0 = 1 + 1 = 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the interval length with the value.

✗ **(C)** Treats sine as odd over a non-symmetric interval.

✗ **(D)** Uses only the upper limit.

Question 50

★★★★★ Very Important

Correct answer: (A) 1

STEP-BY-STEP SOLUTION

An antiderivative is $\ln|x|$. Evaluating gives $\ln e - \ln 1 = 1 - 0 = 1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Assumes the terms cancel.

✗ **(C)** $\ln e$ equals 1, so this is the same value but is not simplified; the fully simplified answer is 1.

✗ **(D)** Confuses the upper limit with the value.

Question 51

★★★ Mandatory

Correct answer: (C) $e - 1$

STEP-BY-STEP SOLUTION

An antiderivative is e^x. Evaluating gives $e^1 - e^0 = e - 1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the lower limit.

✗ **(B)** Forgets to subtract $e^0 = 1$.

✗ **(D)** Adds instead of subtracting.

Question 52

★★★ Mandatory

Correct answer: (A) 36

STEP-BY-STEP SOLUTION

An antiderivative is $x^3/3 - x$. Evaluating gives $(125/3 - 5) - (8/3 - 2) = 110/3 - 2/3 = 108/3 = 36$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses $x^3/3$ only.

✗ **(C)** Forgets to subtract the lower-limit value fully.

✗ **(D)** Arithmetic slip in the subtraction.

Question 53

★★★ Mandatory

Correct answer: (D) 8/3

STEP-BY-STEP SOLUTION

The integrand is even, so the integral is $2\int_0^1(x^2 + 1)\,dx = 2(1/3 + 1) = 8/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes only half of the symmetric interval.

✗ **(B)** Uses only the constant term.

✗ **(C)** Forgets the 1/3 from x^2.

Question 54

★★★ Mandatory

Correct answer: (B) 16/3

STEP-BY-STEP SOLUTION

An antiderivative of x^(1/2) is (2/3)x^(3/2). Evaluating gives (2/3)(8) = 16/3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses x^(3/2) without the 2/3 factor.

✗ **(C)** Uses (1/3) instead of (2/3).

✗ **(D)** Uses the upper limit squared.

Question 55

★★ Important

Correct answer: (A) 10

STEP-BY-STEP SOLUTION

An antiderivative is $x^2 + 3x$. Evaluating gives $(4 + 6) - 0 = 10$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates $x^2 + 3x$ at $x = 2$ incorrectly as 14.

✗ **(C)** Uses only the x^2 term.

✗ **(D)** Uses only $3x$ at $x = 2$ plus 1.

Question 56

★★★ Mandatory

Correct answer: (D) −3

STEP-BY-STEP SOLUTION

An antiderivative is $x^2 - 4x$. Evaluating gives $(9 - 12) - 0 = -3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $-2 \cdot 3$.

✗ **(B)** Assumes the positive and negative areas cancel exactly; they do not.

✗ **(C)** Drops the negative sign.

Question 57

★★★★★ Very Important

Correct answer: (D) F(b) – F(a)

STEP-BY-STEP SOLUTION

The Fundamental Theorem of Calculus gives upper limit minus lower limit: $F(b) - F(a)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reverses the order and gives the negative.

✗ **(B)** Uses the integrand rather than an antiderivative.

✗ **(C)** Differentiates instead of using the antiderivative.

Question 58

★★★ Mandatory

Correct answer: (A) 20/3

STEP-BY-STEP SOLUTION

An antiderivative is $x^3/3 + 2x$. Evaluating gives $8/3 + 4 = 20/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds 2 instead of 4.

✗ **(C)** Forgets the +2 term's contribution of 4.

✗ **(D)** Uses only the $x^3/3$ term at $x = 2$.

Question 59

★★★★★ Very Important

Correct answer: (C) Additivity over intervals

STEP-BY-STEP SOLUTION

Integrals over adjacent intervals add to the integral over the combined interval.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** That concerns inequalities between integrands.

✗ **(B)** That gives $\int_5^0 f = -\int_0^5 f$.

✗ **(D)** That concerns pulling out a constant factor.

Question 60

★★★★★ Very Important

Correct answer: (C) the negative of the value

STEP-BY-STEP SOLUTION

∫_b^a f(x) dx = −∫_a^b f(x) dx.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Not a property of definite integrals.

✗ **(B)** Only true when the integral is zero.

✗ **(D)** Only true if the net area happens to be zero.

Question 61

★★★ Mandatory

Correct answer: (A) $5\int_0^4 f(x)\,dx$

STEP-BY-STEP SOLUTION

A constant factor may be pulled outside the integral.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds inside the integral.

✗ **(C)** Adds the constant instead of multiplying.

✗ **(D)** Multiplies the limits instead of the integrand.

Question 62

★★★★★ Very Important

Correct answer: (C) ∫_a^b f(x) dx ≥ 0

STEP-BY-STEP SOLUTION

A nonnegative integrand produces a nonnegative integral, since it represents signed area above the axis.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Would require $f \le 0$.

✗ **(B)** Only true if f is identically zero.

✗ **(D)** Confuses area with interval length.

Question 63

★★★ Mandatory

Correct answer: (A) 0

STEP-BY-STEP SOLUTION

x^3 is odd and the interval is symmetric about 0, so the integral is 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses 2^3.

✗ **(C)** Computes only part of the interval.

✗ **(D)** Uses 2^4.

Question 64

★★★ Mandatory

Correct answer: (B) 18

STEP-BY-STEP SOLUTION

x^2 is even, so the integral is $2\int_0^3 x^2\,dx = 2(27/3) = 18$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 3^3.

✗ **(C)** Computes only half.

✗ **(D)** Treats x^2 as an odd function.

Question 65

★★★ Mandatory

Correct answer: (A) $\int f \le \int g$

STEP-BY-STEP SOLUTION

The comparison property preserves the inequality after integration.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Reverses the inequality.

✗ **(C)** The comparison property always applies.

✗ **(D)** Equality need not hold.

Question 66

★★ Important

Correct answer: (B) 8

STEP-BY-STEP SOLUTION

The area of a rectangle of height 4 and width 2 is 8.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the width.

✗ **(C)** Adds the height and width.

✗ **(D)** Uses only the height.

Question 67

★★★★★ Very Important

Correct answer: (A) x + 1

STEP-BY-STEP SOLUTION

By the Fundamental Theorem of Calculus, the derivative of an accumulation function with upper limit x is the integrand evaluated at x.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Treats the integrand as constant.

✗ **(C)** This is F(x) itself, not F′(x).

✗ **(D)** Forgets the +1.

Question 68

★★★★★ Very Important

Correct answer: (C) cos x

STEP-BY-STEP SOLUTION

The FTC gives G′(x) = cos x directly.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates the integrand at the lower limit.

✗ **(B)** Antidifferentiates instead of differentiating.

✗ **(D)** Sign error.

Question 69

★★★★★ Very Important

Correct answer: (A) $2x \sin(x^2)$

STEP-BY-STEP SOLUTION

Apply the FTC together with the chain rule: $\sin(x^2) \cdot d/dx(x^2) = 2x \sin(x^2)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the chain rule factor 2x.

✗ **(C)** Differentiates the integrand incorrectly.

✗ **(D)** A derivative does not take a + C.

Question 70

★★★ Mandatory

Correct answer: (A) 12

STEP-BY-STEP SOLUTION

By the FTC, $F'(x) = 3x^2$, so $F'(2) = 12$. The lower limit does not affect the derivative.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses 2^2 instead of $3 \cdot 2^2$.

✗ **(C)** Uses $2 \cdot 2^2$.

✗ **(D)** Assumes the derivative vanishes at the lower limit.

Question 71

★★★ Mandatory

Correct answer: (C) 9

STEP-BY-STEP SOLUTION

$F(x) = x^2$, so $F(3) = 9$. Equivalently, $\int_0^3 2t\,dt = t^2 \big|_0^3 = 9$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to square.

✗ **(B)** Uses $2t$ at $t = 3$.

✗ **(D)** Uses $2 \cdot 3^2$.

Question 72

★★★★★ Very Important

Correct answer: (D) 1

STEP-BY-STEP SOLUTION

$A(e) = \ln|t| \big|_1^e = \ln e - \ln 1 = 1 - 0 = 1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes the integral from a point to itself.

✗ **(B)** Confuses the upper limit with the value.

✗ **(C)** Uses the reciprocal.

Question 73

★★★★★ Very Important

Correct answer: (D) x = −2 and x = 2

STEP-BY-STEP SOLUTION

$F'(x) = x^2 - 4 = (x - 2)(x + 2)$, which is zero at $x = \pm 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the lower limit with a critical number.

✗ **(B)** Misses $x = -2$.

✗ **(C)** F' exists everywhere and has two zeros.

Question 74

★★★★★ Very Important

Correct answer: (C) x < 2

STEP-BY-STEP SOLUTION

$G'(x) = 2 - x$, which is positive exactly when $x < 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is where G' is negative.

✗ **(B)** G' does not depend on the sign of x alone.

✗ **(D)** G' is positive for $x < 2$, including negative x.

Question 75

★★★ Mandatory

Correct answer: (D) 3/2

STEP-BY-STEP SOLUTION

$F(2) = t^2/2 \big|_{-1}^{2} = 4/2 - 1/2 = 3/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the lower-limit contribution.

✗ **(B)** Uses 2^2 without halving.

✗ **(C)** Subtracts incorrectly.

Question 76

★★★★★ Very Important

Correct answer: (A) $3x^2 e^{(x^3)}$

STEP-BY-STEP SOLUTION

FTC plus chain rule: $e^{(x^3)} \cdot d/dx(x^3) = 3x^2 e^{(x^3)}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the chain rule factor.

✗ **(C)** Multiplies by the upper limit instead of its derivative.

✗ **(D)** Misapplies the chain rule to the exponent.

Question 77

★★★★★ Very Important

Correct answer: (C) 3

STEP-BY-STEP SOLUTION

Average value $= (1/3)\int_0^3 x^2\,dx = (1/3)(27/3) = (1/3)(9) = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the upper endpoint value.

✗ **(B)** Forgets to divide by the interval length.

✗ **(D)** Doubles the correct value.

Question 78

★★★ Mandatory

Correct answer: (C) 5

STEP-BY-STEP SOLUTION

Average $= (1/4)\int_0^4 (2x + 1)\,dx = (1/4)(16 + 4) = 20/4 = 5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the upper endpoint value doubled.

✗ **(B)** Forgets to divide by 4.

✗ **(D)** Uses only the x^2 term.

Question 79

★★★★★ Very Important

Correct answer: (D) 2/π

STEP-BY-STEP SOLUTION

Average = $(1/\pi)\int_0^{\pi} \sin x\,dx = (1/\pi)(2) = 2/\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor 2 from the integral.

✗ **(B)** Forgets to divide by π.

✗ **(C)** Assumes symmetry gives zero.

Question 80

★★★ Mandatory

Correct answer: (D) e – 1

STEP-BY-STEP SOLUTION

Average = $(1/1)\int_0^1 e^x\,dx = e - 1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the lower limit.

✗ **(B)** Forgets to subtract $e^0 = 1$.

✗ **(C)** Averages the endpoint values instead of integrating.

Question 81

★★★★★ Very Important

Correct answer: (C) 3

STEP-BY-STEP SOLUTION

Average value = $(1/(6 - 0))\cdot 18 = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the division.

✗ **(B)** Divides by 3 instead of 6.

✗ **(D)** Forgets to divide by the interval length.

Question 82

★★★ Mandatory

Correct answer: (C) 2

STEP-BY-STEP SOLUTION

$\int_0^4 (4 - x)\,dx = 16 - 8 = 8$, and $8/4 = 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes symmetry about zero.

✗ **(B)** Uses half the interval length.

✗ **(D)** Uses only the constant term.

Question 83

★★★★★ Very Important

Correct answer: (B) 8

STEP-BY-STEP SOLUTION

$\int_0^2 3t^2\,dt = t^3 \big|_0^2 = 8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses t^2 at $t = 2$.

✗ **(C)** Uses $3t^2$ at $t = 2$.

✗ **(D)** Uses $3t$ at $t = 2$.

Question 84

★★★★★ Very Important

Correct answer: (C) 12

STEP-BY-STEP SOLUTION

$\int_0^6 (2t - 4)\,dt = [t^2 - 4t]_0^6 = (36 - 24) - 0 = 12$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes the motion cancels.

✗ **(B)** This is the total distance, not the displacement.

✗ **(D)** Forgets the $-4t$ term.

Question 85

★★★★★ Very Important

Correct answer: (B) 20

STEP-BY-STEP SOLUTION

$v = 0$ at $t = 2$. Distance $= \int_0^2 (4 - 2t)\,dt + \int_2^6 (2t - 4)\,dt = 4 + 16 = 20$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds an extra interval.

✗ **(C)** Computes only the second piece plus the first incorrectly.

✗ **(D)** This is the displacement.

Question 86

★★★ Mandatory

Correct answer: (A) 16

STEP-BY-STEP SOLUTION

Distance = rate × time $= 4 \times (5 - 1) = 16$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Halves the interval.

✗ **(C)** Uses time 5 instead of the interval length 4.

✗ **(D)** Uses only the velocity.

Question 87

★★★★★ Very Important

Correct answer: (B) 20

STEP-BY-STEP SOLUTION

$v(3) = v(0) + \int_0^3 6\,dt = 2 + 18 = 20$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the acceleration.

✗ **(C)** Divides by 3 incorrectly.

✗ **(D)** Forgets the initial velocity.

Question 88

★★★★★ Very Important

Correct answer: (D) total distance equals displacement

STEP-BY-STEP SOLUTION

With no sign change, $|v| = v$, so the integrals of v and $|v|$ are equal.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** There is no backtracking, so nothing is added.

✗ **(B)** The particle is moving.

✗ **(C)** Positive velocity gives positive displacement.

Question 89

★★★★★ Very Important

Correct answer: (D) $s(0) + \int_0^t v(x)\,dx$

STEP-BY-STEP SOLUTION

Position is the starting position plus the accumulated change in position.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds the instantaneous velocity rather than the accumulated change.

✗ **(B)** Uses the initial velocity instead of the initial position.

✗ **(C)** Omits the initial position.

Question 90

★★★★★ Very Important

Correct answer: (D) traveled 30 units and ended 12 units from its start

STEP-BY-STEP SOLUTION

The integral of speed is total distance (30); the integral of velocity is net displacement (12).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Swaps the two meanings.

✗ **(B)** Subtracts them, which has no physical meaning here.

✗ **(C)** Adds the two quantities.

Question 91: Riemann Sums from a Table

★★★★★ Very Important

MODEL SOLUTION

(a) $\Delta x = 2$. Left endpoints $x = 0, 2, 4, 6$ give $L_4 = 2(3 + 5 + 4 + 2) = 2(14) = \mathbf{28}$.

(b) Right endpoints $x = 2, 4, 6, 8$ give $R_4 = 2(5 + 4 + 2 + 1) = 2(12) = \mathbf{24}$.

(c) Trapezoid rule: $T_4 = (2/2)[f(0) + 2f(2) + 2f(4) + 2f(6) + f(8)] = 1[3 + 10 + 8 + 4 + 1] = \mathbf{26}$.

This equals $(L_4 + R_4)/2 = (28 + 24)/2 = 26$, because each interior value is counted once in the left sum and once in the right sum, then halved.

SCORING RUBRIC

Criterion	Points
Correct Δx and left endpoints, value 28	3
Correct right endpoints, value 24	3
Correct trapezoid setup and value 26 with explanation	3
Total	9

Question 92: Definite Integrals from a Graph

★★★★★ Very Important

MODEL SOLUTION

(a) Areas by region: [0,2] triangle above the axis, area 4, integral +4. The segment from (2,4) to (5,−2) crosses zero at t = 4: above-axis triangle on [2,4] has area 4, below-axis triangle on [4,5] has area 1, contributing −1. On [5,8] the triangle lies below the axis with area 3, contributing −3.
So $\int_0^8 f(t)\,dt = 4 + 4 - 1 - 3 = \mathbf{4}$.
(b) Total area uses absolute values: 4 + 4 + 1 + 3 = **12**.
(c) Average value = $(1/8)\cdot\int_0^8 f(t)\,dt = 4/8 = \mathbf{1/2}$.
(d) Part (a) is the *net signed* area, where regions below the axis subtract. Part (b) is the *total* area, where every region is counted positively.

SCORING RUBRIC

Criterion	Points
Correct region-by-region signed areas	1
Correct net integral value 4	2
Correct total area 12	3
Correct average value 1/2	2
Explains signed versus total area	1
Total	9

Question 93: Analyzing an Accumulation Function

★★★★★ Very Important

MODEL SOLUTION

(a) By the Fundamental Theorem of Calculus, $\mathbf{F'(x) = x^2 - 4}$.

(b) $F(x) = t^3/3 - 4t\,|_0^x = \mathbf{x^3/3 - 4x}$.

(c) $F'(x) = (x - 2)(x + 2) = 0$ at $x = \pm 2$. F' is positive for $x < -2$, negative for $-2 < x < 2$, and positive for $x > 2$. So $x = -2$ is a **local maximum** and $x = 2$ is a **local minimum**.

(d) $F(2) = 8/3 - 8 = \mathbf{-16/3}$. $F(-2) = -8/3 + 8 = \mathbf{16/3}$.

SCORING RUBRIC

Criterion	Points
States $F'(x) = x^2 - 4$ with the FTC	2
Correct antiderivative formula	3
Both critical numbers found	1
Correct max/min classification	1
Both values $-16/3$ and $16/3$	2
Total	9

Question 94: FTC and Average Value

★★★★★ Very Important

MODEL SOLUTION

(a) An antiderivative is $x^3 - x^2$. So $\int_1^4 f = (64 - 16) - (1 - 1) = 48 - 0 = \mathbf{48}$.

(b) Average value = $48/(4 - 1) = \mathbf{16}$.

(c) Solve $3c^2 - 2c = 16$, that is $3c^2 - 2c - 16 = 0$.

$c = [2 \pm \sqrt{(4 + 192)}]/6 = [2 \pm 14]/6$, so $c = 8/3$ or $c = -2$.

Only $c = \mathbf{8/3}$ lies in $[1, 4]$.

SCORING RUBRIC

Criterion	Points
Correct antiderivative	1
Correct evaluation 48	2
Average value 16	2
Sets up $3c^2 - 2c = 16$	2
Selects $c = 8/3$ and rejects -2	2
Total	9

Question 95: Motion: Displacement and Distance

★★★★★ Very Important

MODEL SOLUTION

(a) Displacement = $\int_0^5 (t^2 - 4t + 3)\,dt = [t^3/3 - 2t^2 + 3t]_0^5 = 125/3 - 50 + 15 = 125/3 - 35 = \mathbf{20/3\ m}$.

(b) $v(t) = (t - 1)(t - 3)$ is zero at $t = 1$ and $t = 3$.

$\int_0^1 v = 1/3 - 2 + 3 = 4/3$. $\int_1^3 v = 0 - 4/3 = -4/3$. $\int_3^5 v = 20/3 - 0 = 20/3$.

Total distance = $|4/3| + |-4/3| + |20/3| = 4/3 + 4/3 + 20/3 = \mathbf{28/3\ m}$.

(c) $v(t) > 0$ on **(0, 1) and (3, 5]**.

SCORING RUBRIC

Criterion	Points
Correct antiderivative and displacement 20/3	3
Identifies sign-change times t = 1 and t = 3	1
Splits the integral into three pieces	1
Correct total distance 28/3	2
Correct positive-direction intervals	2
Total	9

Question 96: From a Riemann Sum to an Integral

★★★★★ Very Important

MODEL SOLUTION

(a) $\Delta x = 6/n$. The right endpoints are $x_i = a + i\Delta x = 6i/n$, so $a = 0$ and $b = 6$. The interval is **[0, 6]**.

(b) Since the sampled height is $(2 + x_i)^2$, the integrand is $f(x) = (2 + x)^2$ and the limit equals $\int_0^6 \mathbf{(2 + x)^2\,dx}$.

(c) $\int_0^6 (2 + x)^2\,dx = [(2 + x)^3/3]_0^6 = 512/3 - 8/3 = 504/3 = \mathbf{168}$.

Check by expanding: $\int_0^6 (x^2 + 4x + 4)\,dx = 72 + 72 + 24 = 168$.

SCORING RUBRIC

Criterion	Points
Identifies $\Delta x = 6/n$	1
Identifies $x_i = 6i/n$ and interval [0,6]	2
Writes the integral with $f(x) = (2+x)^2$	3
Antiderivative $(2+x)^3/3$	1
Exact value 168	2
Total	9

Question 97: Integral of a Piecewise Function

★★★ Mandatory

MODEL SOLUTION

(a) Split at $x = 2$.

$\int_0^2 (x + 1)\,dx = [x^2/2 + x]_0^2 = 2 + 2 = 4$.

$\int_2^5 (5 - x)\,dx = [5x - x^2/2]_2^5 = (25 - 25/2) - (10 - 2) = 12.5 - 8 = 4.5$.

Total $= 4 + 4.5 =$ **17/2**.

(b) Average value $= (17/2)/5 =$ **17/10**.

(c) Try the first piece: $x + 1 = 17/10$ gives $x = 7/10 = 0.7$, which lies in $[0, 2]$.

So **c = 7/10** works.

SCORING RUBRIC

Criterion	Points
Correct split and first piece value 4	1
Correct second piece value 9/2	1
Correct total 17/2	1
Average value 17/10	2
Solves on the correct piece and finds c = 7/10	2
Verifies c lies in [0,2]	2
Total	9

Question 98: Accumulation with a Variable Upper Limit

★★★★★ Very Important

MODEL SOLUTION

(a) By the FTC and chain rule, $G'(x) = (x^2 + 1)\cdot 2x = \mathbf{2x^3 + 2x}$.

(b) $G(x) = [t^2/2 + t]_1^{(x^2)} = (x^4/2 + x^2) - (1/2 + 1) = x^4/2 + x^2 - 3/2$.
Differentiating gives $2x^3 + 2x$, which confirms part (a).

(c) $G(2) = 16/2 + 4 - 3/2 = 8 + 4 - 3/2 = \mathbf{21/2}$.

(d) $G'(x) = 2x(x^2 + 1) = 0$ only at $\mathbf{x = 0}$, since $x^2 + 1 > 0$ always.

SCORING RUBRIC

Criterion	Points
Applies FTC and chain rule correctly	2
Correct $G'(x) = 2x^3 + 2x$	1
Correct explicit formula and confirms by differentiating	2
$G(2) = 21/2$	2
Identifies $x = 0$ as the only critical number	2
Total	9

Question 99: Applied Accumulation from a Table

★★★★★ Very Important

MODEL SOLUTION

(a) $\Delta t = 2$. $T_4 = (2/2)[R(0) + 2R(2) + 2R(4) + 2R(6) + R(8)]$
$= 1[5 + 14 + 12 + 6 + 2] =$ **39 liters**.
(b) $L_4 = 2(5 + 7 + 6 + 3) = 2(21) =$ **42 liters**.
$R_4 = 2(7 + 6 + 3 + 2) = 2(18) =$ **36 liters**.
(c) $(L_4 + R_4)/2 = (42 + 36)/2 = 39$, which equals T_4.
(d) Approximately **39 liters** of water entered the tank during the first 8 minutes.

SCORING RUBRIC

Criterion	Points
Correct trapezoid setup	1
Correct value 39	2
Correct left sum 42	1
Correct right sum 36	1
Shows the averaging relationship	2
Interprets 39 liters over 8 minutes	2
Total	9

Question 100: Net Area versus Total Area

★★★★★ Very Important

MODEL SOLUTION

(a) $\int_0^5 (x - 2)\,dx = [x^2/2 - 2x]_0^5 = (25/2 - 10) - 0 = \mathbf{5/2}$.

(b) $f(x) = 0$ at $x = 2$. On $[0, 2]$, $f \le 0$; on $[2, 5]$, $f \ge 0$.

$\int_0^2 (2 - x)\,dx = [2x - x^2/2]_0^2 = 4 - 2 = 2$.

$\int_2^5 (x - 2)\,dx = [x^2/2 - 2x]_2^5 = (25/2 - 10) - (2 - 4) = 5/2 + 2 = 9/2$.

Total area $= 2 + 9/2 = \mathbf{13/2}$.

(c) Average value $= (1/5)(5/2) = \mathbf{1/2}$.

(d) Part (a) is net signed area: the region below the axis on $[0, 2]$ is subtracted. Part (b) counts that same region as positive area, so it is larger.

SCORING RUBRIC

Criterion	Points
Correct integral 5/2	2
Identifies zero at x = 2 and splits integral	1
Evaluates piece areas	1
Correct total area 13/2	1
Average value 1/2	2
Explains signed versus total area	2
Total	9

Essential Unit 6 Facts

- **Net change:** if f′ is a rate, then f(b) = f(a) + ∫_a^b f′(t) dt.
- **FTC:** if F′ = f, then ∫_a^b f(x) dx = F(b) − F(a).
- **Accumulation function:** d/dx [∫_a^x f(t) dt] = f(x); with upper limit u(x), multiply by u′(x).
- **Riemann sums:** left, right, and midpoint sums converge to the definite integral as n → ∞.
- **Trapezoid rule:** T_n = (Δx/2)[f(x₀) + 2f(x₁) + … + 2f(x_{n−1}) + f(x_n)]; it equals (L_n + R_n)/2.
- **Integral properties:** additivity over intervals, reversing limits changes sign, constant factors come out.
- **Motion:** displacement = ∫v dt; total distance = ∫|v| dt; speed increases when v and a share a sign.
- **Average value:** f_avg = (1/(b − a))∫_a^b f(x) dx.
- **Even/odd:** over [−a, a], an even integrand doubles the half-interval integral and an odd integrand gives 0.

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